

Maharashtra Public Service Commission Assessment

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Assistant Professor Metallurgy Maharashtra Engineering Teachers Service Group A

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Assistant Professor Metallurgy Maharashtra Engineering Teachers Service Group A

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Question Number : 1 Question Id : 4410092079630 Question Type : MCQ Option Shuffling : No

Based on the Gibbs relation between standard Gibbs free energy change and equilibrium constant, what happens to the equilibrium constant (K) when the temperature of an exothermic reaction is increased?

Options :

1. The equilibrium constant is undefined
2. The equilibrium constant decreases
3. The equilibrium constant remains unchanged
4. The equilibrium constant increases

Question Number : 2 Question Id : 4410092079633 Question Type : MCQ Option Shuffling : No

Why does the line representing the reaction $2\text{C(s)} + \text{O}_2\text{(g)} = 2\text{CO(g)}$ have a negative slope in an Ellingham diagram?

Options :

1. The reaction is highly exothermic at all temperatures, resulting in a negative enthalpy change
2. The reaction produces an equal volume of gas as consumed, keeping the entropy change zero
3. The reaction consumes oxygen gas to form a solid product, resulting in a negative entropy change
4. The reaction results in an increase in the number of moles of gaseous products, leading to a positive entropy change

Question Number : 3 Question Id : 4410092079652 Question Type : MCQ Option Shuffling : No

In the context of oxidation kinetics, which of the following conditions is most likely to lead to the linear rate law dominating the oxidation of a metal surface?

Options :

1. The oxide film that forms is porous and non-protective
2. The oxide film forms a dense, continuous barrier
3. The oxide film is glassy and impermeable
4. The oxide film is adherent and continuous

Question Number : 4 Question Id : 4410092079628 Question Type : MCQ Option Shuffling : No

In a binary alloy system at constant temperature, the common tangent to the free energy curves of two phases indicates which of the following?

Options :

1. The fraction of each phase present in the mixture
2. The pressure at which the phases are stable
3. The temperature at which two phases coexist
4. The equilibrium compositions of the two coexisting phases

Question Number : 5 Question Id : 4410092079635 Question Type : MCQ Option Shuffling : No

An engineer is troubleshooting corrosion in a steel plant. Based on the Fe-S-O predominance diagram, which change would most likely shift the stability region from iron oxide to iron sulfide at constant temperature?

Options :

1. Increasing oxygen partial pressure while keeping sulfur constant
2. Decreasing sulfur partial pressure with constant oxygen level
3. Decreasing oxygen partial pressure and increasing sulfur partial pressure
4. Keeping both oxygen and sulfur partial pressures constant

Question Number : 6 Question Id : 4410092079640 Question Type : MCQ Option Shuffling : No

A metallurgical engineer is analyzing the feasibility of using a copper electrode in an electrochemical cell with a zinc electrode. If the standard reduction potentials are $E^\circ(\text{Cu}^{2+}/\text{Cu}) = +0.34 \text{ V}$ and $E^\circ(\text{Zn}^{2+}/\text{Zn}) = -0.76 \text{ V}$, which of the following statements correctly describes the role of copper in this cell under standard conditions?

Options :

1. Zinc will be reduced at the copper electrode.
2. Copper will be oxidized while zinc is reduced at the copper electrode.
3. Copper will function as the anode and undergo oxidation.
4. Copper will function as the cathode and undergo reduction.

Question Number : 7 Question Id : 4410092079649 Question Type : MCQ Option Shuffling : No

A metallurgical engineer is studying the reduction kinetics of a spherical metal oxide pellet in a packed bed. Experimental data reveals that after an initial rapid conversion, the reaction rate slows drastically and is inversely proportional to the thickness of the newly formed solid metal shell. According to the shrinking core model, which mechanism predominantly controls the reaction rate during this slower phase?

Options :

1. Diffusion through the product layer
2. Pore diffusion through the unreacted core
3. Gas film resistance at the boundary layer
4. Chemical reaction kinetics at the shrinking interface

Question Number : 8 Question Id : 4410092079650 Question Type : MCQ Option Shuffling : No

During a high-temperature gas carburizing process, the mass transfer coefficient of carbon at the gas-solid interface is orders of magnitude greater than the rate of interstitial mass transport within the solid lattice. Which factor primarily limits the rate of case hardening?

Options :

1. The composition of the furnace atmosphere
2. The diffusion coefficient of carbon in steel
3. The surface area of the steel component
4. The temperature of the furnace

Question Number : 9 Question Id : 4410092079611 Question Type : MCQ Option Shuffling : No

A metallurgical reactor undergoes a cyclic process where heat is absorbed from a high-temperature reservoir and rejected to a low-temperature reservoir. The sum of all $\left(\frac{\delta Q}{T}\right)$ for the system over one cycle is found to be zero. What conclusion can be drawn about the process?

Options :

1. The cyclic process is reversible and adheres to the Clausius inequality
2. The process is impossible because it violates the Second Law
3. The system exchanges heat at only one temperature during the cycle
4. The process is irreversible due to entropy generation in the surroundings

Question Number : 10 Question Id : 4410092079612 Question Type : MCQ Option Shuffling : No

An engineering alloy undergoes a phase transformation at 700 K. The enthalpy change (ΔH) for the transformation is 10 kJ/mol, and the entropy change (ΔS) is 20 J/mol·K. What is the Gibbs free energy change (ΔG) for this transformation at 700 K?

Options :

1. 4 kJ/mol
2. -24 kJ/mol
3. -4 kJ/mol
4. 24 kJ/mol

Question Number : 11 Question Id : 4410092079613 Question Type : MCQ Option Shuffling : No

During the design of a thermodynamic engine, a reversible isothermal process at 400 K and constant volume involves decreasing the system's entropy by 3 J/K. What is the minimum work that must be supplied?

Options :

1. 800 J
2. -1200 J
3. -800 J
4. 1200 J

Question Number : 12 Question Id : 4410092079616 Question Type : MCQ Option Shuffling : No

A mechanical engineer is designing a lubrication system using an ideal solution of two lubricants, X and Y. At 350 K, the vapor pressures of pure X and Y are 200 mmHg and 150 mmHg, respectively. If the mole fraction of X in the solution is 0.4, what is the total vapor pressure over the solution?

Options :

1. 180 mmHg
2. 165 mmHg
3. 155 mmHg
4. 170 mmHg

Question Number : 13 Question Id : 4410092079618 Question Type : MCQ Option Shuffling : No

During the design of a high-temperature reactor, a metallurgical engineer observes that the measured activity of oxygen in a molten iron solution is 0.012 at a mole fraction of 0.008. Assuming that the liquid mixture behaves as a regular solution, what is the predicted Raoultian activity of oxygen (a_o) when the composition is altered to a mole fraction of 0.025, and how is the solution's deviation from ideal Raoultian behavior characterized?

Options :

1. 0.0169; Negative deviation
2. 0.0250; Ideal behavior
3. 0.0370; Positive deviation
4. 0.0412; Positive deviation

Question Number : 14 Question Id : 4410092079625 Question Type : MCQ Option Shuffling : No

An engineer analyzes a binary alloy system displaying a peritectic reaction at 780°C. If the cooling rate is faster than equilibrium conditions, which of the following is the microstructural outcome at the peritectic composition?

Options :

1. Incomplete peritectic transformation due to limited diffusion
2. Formation of pure peritectic phase without remnants
3. Complete disappearance of primary phase before peritectic temperature
4. Direct eutectic structure formation at peritectic temperature

Question Number : 15 Question Id : 4410092079642 Question Type : MCQ Option Shuffling : No

A mechanical engineer is designing a galvanic sensor using a silver electrode in contact with a 0.001 M AgNO_3 solution. The standard reduction potential for Ag^+/Ag is +0.80 V. Calculate the actual electrode potential at 25 degrees Celsius. (Use the Nernst equation and assume 1 electron transfer.)

Options :

1. 0.74 V
2. 0.62 V
3. 0.56 V
4. 0.80 V

Question Number : 16 Question Id : 4410092079645 Question Type : MCQ Option Shuffling : No

During a steel carburization process, the measured carbon concentration profile in the solid reveals a steep gradient near the surface and a much shallower gradient deeper inside. Which kinetic regime most likely governs the rate law in this case?

Options :

1. Diffusion-controlled regime
2. Mixed kinetic regime
3. Equilibrium regime
4. Interface-controlled regime

Question Number : 17 Question Id : 4410092079661 Question Type : MCQ Option Shuffling : No

Which dimensionless number is most relevant for describing the flow regime when a viscous fluid flows past a flat plate at low velocity?

Options :

1. Prandtl number
2. Reynolds number
3. Nusselt number
4. Schmidt number

Question Number : 18 Question Id : 4410092079667 Question Type : MCQ Option Shuffling : No

If the thermal conductivity of a plane wall doubles while all other parameters remain unchanged, how does the steady-state one-dimensional heat transfer rate change?

Options :

1. It is reduced by half
2. It becomes four times as large
3. It remains unchanged
4. It becomes twice as large

Question Number : 19 Question Id : 4410092079671 Question Type : MCQ Option Shuffling : No

A flat plate is cooled by air flowing over it. If the velocity of air increases, what happens to the heat transfer coefficient, assuming other factors remain constant?

Options :

1. The heat transfer coefficient increases
2. The heat transfer coefficient decreases
3. The heat transfer coefficient fluctuates
4. The heat transfer coefficient remains unchanged

Question Number : 20 Question Id : 4410092079672 Question Type : MCQ Option Shuffling : No

In a heat exchanger, oil is pumped through a tube at a constant rate while water flows outside the tube with forced convection. If the oil's viscosity decreases with temperature, what is the impact on the heat transfer rate over time?

Options :

1. The heat transfer rate increases as oil viscosity decreases
2. The heat transfer rate remains unchanged regardless of viscosity
3. The effect on heat transfer rate is negligible over time
4. The heat transfer rate decreases with lower oil viscosity

Question Number : 21 Question Id : 4410092079681 Question Type : MCQ Option Shuffling : No

Which physical law relates the total energy radiated per unit area by a black body to its absolute temperature raised to the fourth power?

Options :

1. Kirchhoff's Law
2. Planck's Law
3. Stefan Boltzmann Law
4. Wien's Law

Question Number : 22 Question Id : 4410092079683 Question Type : MCQ Option Shuffling : No

Which scenario best demonstrates Kirchhoff's Law during a metallurgical process?

Options :

1. A steel plate with poor absorption emits more heat than a good absorber
2. A steel plate with low absorptivity radiates heat efficiently at any temperature
3. A steel plate with high absorptivity also radiates heat efficiently at the same temperature
4. A steel plate with high absorptivity radiates heat poorly at the same temperature

Question Number : 23 Question Id : 4410092079693 Question Type : MCQ Option Shuffling : No

A fluid flow is characterized by a high Reynolds number. What does this indicate about the relative importance of inertial and viscous forces?

Options :

1. Viscous forces dominate inertial forces
2. Inertial forces dominate viscous forces
3. Inertial and viscous forces are equal
4. Neither inertial nor viscous forces are significant

Question Number : 24 Question Id : 4410092079659 Question Type : MCQ Option Shuffling : No

A process engineer is tasked with minimizing the pumping power required to transport oil through a pipeline over a certain distance. Which modification would most directly reduce the head loss term in the mechanical energy balance equation without altering the flow rate or pipe diameter?

Options :

1. Increasing the velocity of the fluid
2. Increasing the elevation of the pipeline
3. Using a denser fluid for transport
4. Reducing the pipe surface roughness

Question Number : 25 Question Id : 4410092079664 Question Type : MCQ Option Shuffling : No

A steel rod of length 0.5 m and cross-sectional area 0.01 m^2 is insulated along its length except at the ends. One end is maintained at 400 K and the other at 300 K. If the thermal conductivity is 50 W/mK, what is the steady-state heat transfer rate through the rod?

Options :

1. 150 W
2. 50 W
3. 200 W
4. 100 W

Question Number : 26 Question Id : 4410092079678 Question Type : MCQ Option Shuffling : No

Which of the following quantities is not used in the definition of the Nusselt number for a flat plate?

Options :

1. Thermal conductivity of the fluid
2. Characteristic length of the plate
3. Fluid density near the wall
4. Convective heat transfer coefficient

Question Number : 27 Question Id : 4410092079682 Question Type : MCQ Option Shuffling : No

If the temperature of a black body doubles, by what factor does the energy radiated per unit area increase according to the Stefan Boltzmann Law?

Options :

1. Two
2. Four
3. Sixteen
4. Eight

Question Number : 28 Question Id : 4410092079688 Question Type : MCQ Option Shuffling : No

During carburization of a steel component, a metallurgical engineer measures a steady-state carbon flux of $4 \times 10^{-8} \text{ mol} \cdot \text{m}^{-2} \cdot \text{s}^{-1}$. The concentration gradient across the diffusion zone is determined to be $-2 \times 10^{-3} \text{ mol} \cdot \text{m}^{-4}$. Determine the diffusion coefficient of carbon in the steel.

Options :

1. $2.0 \times 10^{-11} \frac{\text{m}^2}{\text{s}}$
2. $2.0 \times 10^{-5} \frac{\text{m}^2}{\text{s}}$
3. $8.0 \times 10^{-11} \frac{\text{m}^2}{\text{s}}$
4. $8.0 \times 10^{-5} \frac{\text{m}^2}{\text{s}}$

Question Number : 29 Question Id : 4410092079694 Question Type : MCQ Option Shuffling : No

Which dimensionless number is most suitable for characterizing mass transfer when both viscosity and mass diffusivity are important?

Options :

1. Reynolds number
2. Schmidt number
3. Sherwood number
4. Prandtl number

Question Number : 30 Question Id : 4410092079696 Question Type : MCQ Option Shuffling : No

During performance evaluation of an electrochemical cell, the discharge current is progressively increased while temperature and electrolyte composition are kept constant. The measured terminal voltage decreases by an amount greater than that predicted from the equilibrium potential, and the deviation becomes more pronounced at higher current densities. Which electrochemical phenomenon best explains this behaviour?

Options :

1. Electrode polarization
2. Electrolyte conductivity
3. Electrode capacitance
4. Current efficiency

Question Number : 31 Question Id : 4410092079698 Question Type : MCQ Option Shuffling : No

Which equipment is commonly used for primary crushing of large ore pieces in mineral processing plants?

Options :

1. Ball mill
2. Jaw crusher
3. Rod mill
4. Cone crusher

Question Number : 32 Question Id : 4410092079703 Question Type : MCQ Option Shuffling : No

In the sintering process, what is typically used to bind fine ore particles together?

Options :

1. Heat
2. Pressure
3. Water
4. Acid

Question Number : 33 Question Id : 4410092079713 Question Type : MCQ Option Shuffling : No

What metal is commonly extracted using electrometallurgy from its molten salt?

Options :

1. Iron
2. Aluminium
3. Copper
4. Zinc

Question Number : 34 Question Id : 4410092079717 Question Type : MCQ Option Shuffling : No

Which property is most important for metallurgical coke used in blast furnaces?

Options :

1. High moisture content
2. High mechanical strength
3. Low ash content
4. High sulfur content

Question Number : 35 Question Id : 4410092079726 Question Type : MCQ Option Shuffling : No

During ladle refining, an element is added that forms a highly stable oxide and thereby suppresses blowholes during steel solidification. Which element is most commonly used?

Options :

1. Silicon
2. Nitrogen
3. Aluminum
4. Manganese

Question Number : 36 Question Id : 4410092079732 Question Type : MCQ Option Shuffling : No

How does inclusion control benefit the continuous casting process in steel plants?

Options :

1. It reduces the risk of nozzle clogging
2. It improves desulphurization efficiency during ladle treatment
3. It enhances deoxidation reaction rates in molten steel
4. It increases slag basicity during secondary refining

Question Number : 37 Question Id : 4410092079736 Question Type : MCQ Option Shuffling : No

Why is it important for the mold in continuous casting to have good contact with the cooling water?

Options :

1. To promote desulphurization of molten steel
2. To ensure efficient heat transfer
3. To increase inclusion flotation in the mold region
4. To improve alloy homogenization during casting

Question Number : 38 Question Id : 4410092079699 Question Type : MCQ Option Shuffling : No

A concentrator processes two ore samples in the same ball mill under identical operating conditions (mill speed, ball charge, circulating load, and target product size). Sample A requires significantly higher specific grinding energy than Sample B to achieve the same product size distribution. Which feed property most directly explains the higher energy requirement?

Options :

1. Higher Bond Work Index of Sample A
2. Higher bulk density of Sample A
3. Lower moisture content of Sample A
4. Higher pulp density during classification

Question Number : 39 Question Id : 4410092079701 Question Type : MCQ Option Shuffling : No

During the flotation of a sulfide ore, the operator observes that mineral particles attach to air bubbles, but the froth at the top of the flotation cell collapses rapidly before it can be removed as concentrate. Which modification would most directly improve concentrate recovery?

Options :

1. Increase the wettability of the valuable mineral particles
2. Stabilize the froth by adding a suitable frother
3. Dissolve the valuable mineral particles into the pulp
4. Prevent the formation of air bubbles in the cell

Question Number : 40 Question Id : 4410092079702 Question Type : MCQ Option Shuffling : No

A mineral processing plant treats an ore containing valuable heavy minerals and lighter gangue of similar particle sizes. The process engineer intends to exploit differences in specific gravity under pulsating fluid conditions to achieve separation before further concentration. Which equipment would be the most appropriate choice?

Options :

1. Jig
2. Magnetic separator
3. Froth flotation cell
4. Ball mill

Question Number : 41 Question Id : 4410092079723 Question Type : MCQ Option Shuffling : No

Which refractory material is most commonly used as the lining of a modern basic Electric Arc Furnace (EAF) for primary steelmaking?

Options :

1. Fireclay bricks
2. Alumina bricks
3. Silica bricks
4. Magnesite bricks

Question Number : 42 Question Id : 4410092079727 Question Type : MCQ Option Shuffling : No

A steelmaker intends to produce rimmed steel in which vigorous carbon monoxide evolution occurs during solidification. Which deoxidation practice is most appropriate?

Options :

1. Complete aluminium deoxidation before casting
2. Silicon–manganese deoxidation before casting
3. Little or no intentional deoxidation before casting
4. Vacuum deoxidation followed by calcium treatment

Question Number : 43 Question Id : 4410092079729 Question Type : MCQ Option Shuffling : No

During ladle refining, a steelmaker observes significant variations in temperature and alloy composition between samples taken from different locations within the ladle. Which secondary steelmaking operation would most effectively minimize these variations and improve process consistency?

Options :

1. Vacuum degassing
2. Argon stirring
3. Oxygen blowing
4. Slag skimming

Question Number : 44 Question Id : 4410092079733 Question Type : MCQ Option Shuffling : No

A ladle contains 180 tonnes of steel with an inclusion content of 50 ppm. After refining, the inclusion content decreases to 15 ppm. Assuming 1 ppm = 1 g inclusion per tonne, how many grams of inclusions have been removed?

Options :

1. 4500 g
2. 5400 g
3. 6300 g
4. 7200 g

Question Number : 45 Question Id : 4410092079740 Question Type : MCQ Option Shuffling : No

What is the primary characteristic of metallic bonding?

Options :

1. Fixed positions for all electrons in the structure
2. Electron pairs shared between two atoms
3. Complete transfer of electrons from one atom to another
4. Delocalized electrons around positive ions

Question Number : 46 Question Id : 4410092079748 Question Type : MCQ Option Shuffling : No

Which one of the following metals typically exhibits a face centered cubic (FCC) crystal structure at room temperature?

Options :

1. Beryllium
2. Zinc
3. Aluminum
4. Magnesium

Question Number : 47 Question Id : 4410092079758 Question Type : MCQ Option Shuffling : No

Which of the following best describes a surface defect in a crystalline material?

Options :

1. An imperfection that occurs at the boundary of a crystal
2. An imperfection found inside the crystal lattice
3. A region with a change in atomic arrangement throughout the bulk
4. A point where two different phases meet in a material

Question Number : 48 Question Id : 4410092079771 Question Type : MCQ Option Shuffling : No

Which factor most significantly influences the secondary arm spacing during dendritic solidification in alloys?

Options :

1. Cooling rate during solidification
2. Grain boundary migration speed
3. Presence of inert gas in furnace
4. Density of the liquid metal

Question Number : 49 Question Id : 4410092079778 Question Type : MCQ Option Shuffling : No

Which phase transformation does the nose of a typical time-temperature-transformation (TTT) diagram for eutectoid steel represent?

Options :

1. The fastest initiation of austenite transforming to pearlite
2. The slow transformation of austenite to ferrite
3. The decomposition of pearlite to bainite
4. The precipitation of carbides from bainite

Question Number : 50 Question Id : 4410092079781 Question Type : MCQ Option Shuffling : No

In an industrial heat treatment process, why is annealing performed after significant cold work?

Options :

1. To increase hardness by grain refinement
2. To induce martensitic transformation
3. To restore ductility by recrystallization
4. To improve surface roughness

Question Number : 51 Question Id : 4410092079783 Question Type : MCQ Option Shuffling : No

Which surface hardening method is most suitable for achieving a high-carbon martensite layer without affecting the core of a low-carbon steel part?

Options :

1. Annealing
2. Tempering
3. Spheroidizing
4. Carburizing

Question Number : 52 Question Id : 4410092079788 Question Type : MCQ Option Shuffling : No

Which of the following best explains why galvanic corrosion occurs when two dissimilar metals are electrically connected in an electrolyte?

Options :

1. Presence of an electrolyte prevents corrosion between dissimilar metals
2. Both metals corrode at the same rate regardless of their electrode potentials
3. The metal with the more negative electrode potential acts as the anode and corrodes faster
4. Electrons flow from the metal with a positive potential to the one with a negative potential, causing uniform corrosion

Question Number : 53 Question Id : 4410092079739 Question Type : MCQ Option Shuffling : No

Nitrogen gas (N_2) is significantly less reactive at room temperature than many other diatomic molecules. Which property of the N_2 molecule is primarily responsible for its high chemical stability?

Options :

1. Transfer of electrons from one nitrogen atom to the other
2. Presence of a triple covalent bond with very high bond dissociation energy
3. Formation of an extended lattice structure between neighboring molecules
4. Attraction of free electrons from surrounding molecules

Question Number : 54 Question Id : 4410092079741 Question Type : MCQ Option Shuffling : No

Which type of bonding is responsible for holding together the layers in graphite?

Options :

1. Ionic bond
2. Van der Waals force
3. Hydrogen bond
4. Covalent bond

Question Number : 55 Question Id : 4410092079747 Question Type : MCQ Option Shuffling : No

During the heat treatment of a body centered cubic (BCC) alloy, a graduate engineer observes a change in hardness. Which primary characteristic of BCC structure explains this behavior?

Options :

1. BCC structures have more open spaces affecting diffusion rates
2. BCC unit cells possess maximum atomic packing factor among metals
3. BCC lattices have higher atomic coordination numbers reducing defects
4. BCC metals feature close-packed planes increasing dislocation mobility

Question Number : 56 Question Id : 4410092079752 Question Type : MCQ Option Shuffling : No

Which practical challenge can most significantly influence the accuracy of lattice parameter determination using Bragg's Law in an industrial powder X-Ray diffraction (XRD) setup?

Options :

1. Variation in particle size of typical powder sample
2. Machine based automated peak identification for sample
3. Incorrect sample mass for the holder
4. Sample displacement causing peak position errors

Question Number : 57 Question Id : 4410092079755 Question Type : MCQ Option Shuffling : No

Which factor most critically affects the spatial resolution limit in conventional Transmission Electron Microscopy (TEM) imaging of metallic alloys?

Options :

1. Thickness of the specimen foil
2. Voltage stability of the electron gun
3. Magnification setting of the projector lens
4. Spherical aberration of electromagnetic lenses

Question Number : 58 Question Id : 4410092079757 Question Type : MCQ Option Shuffling : No

In evaluating a failed steel component, it is observed that slip bands are aligned along specific planes. Which aspect of line defects would you prioritize to analyze the root cause of failure?

Options :

1. Examine the dislocation movement along slip planes under applied stress
2. Quantify the vacancy concentration through positron annihilation
3. Measure the average grain boundary angles in the microstructure
4. Assess the diffusion rate of interstitial atoms in the lattice

Question Number : 59 Question Id : 4410092079759 Question Type : MCQ Option Shuffling : No

When applying Fick's first law in an industrial process, why is steady-state diffusion a critical assumption?

Options :

1. It makes the flux independent of the material properties
2. It guarantees complete mixing of the solute in the solvent
3. It ensures the diffusion coefficient does not change with temperature
4. It allows the concentration gradient to remain constant over time

Question Number : 60 Question Id : 4410092079760 Question Type : MCQ Option Shuffling : No

Which condition must be satisfied for Fick's Second Law to be applicable during the carburizing process of steel at elevated temperatures?

Options :

1. The concentration gradient must change with time at each position
2. The diffusion coefficient must be zero at all temperatures
3. The concentration gradient must remain constant at each position
4. The solute atoms must not interact with the solvent atoms

Question Number : 61 Question Id : 4410092079797 Question Type : MCQ Option Shuffling : No

Which type of deformation occurs after yielding and is non-reversible?

Options :

1. Plastic deformation
2. Thermal deformation
3. Creep deformation
4. Elastic deformation

Question Number : 62 Question Id : 4410092079825 Question Type : MCQ Option Shuffling : No

Which of the following is considered the rate-controlling process for creep in metals at intermediate temperatures and moderate stresses?

Options :

1. Mechanical twinning of crystals
2. Grain boundary sliding between grains
3. Lattice diffusion of atoms
4. Dislocation climb over obstacles

Question Number : 63 Question Id : 4410092079830 Question Type : MCQ Option Shuffling : No

Which formula correctly represents the volume V of penetrant absorbed in a surface-breaking crack of width w, depth d, and length l?

Options :

1. $V = w + d + l$
2. $V = (w \times d) + l$
3. $V = l \times w/d$
4. $V = w \times d \times l$

Question Number : 64 Question Id : 4410092079809 Question Type : MCQ Option Shuffling : No

According to the Hall-Petch relationship, how does the yield strength of a metal change with decreasing average grain diameter?

Options :

1. Yield strength remains unaffected by grain diameter
2. Yield strength increases with decreasing grain diameter
3. Yield strength becomes unpredictable with grain diameter
4. Yield strength decreases with decreasing grain diameter

Question Number : 65 Question Id : 4410092079811 Question Type : MCQ Option Shuffling : No

Which primary assumption is made in the Griffith theory regarding crack propagation in brittle materials?

Options :

1. The material contains pre-existing microscopic cracks
2. The material is completely free of defects
3. The material fails only under compressive stress
4. The material is perfectly ductile

Question Number : 66 Question Id : 4410092079815 Question Type : MCQ Option Shuffling : No

Which condition is most likely to promote brittle fracture in steel components?

Options :

1. High strain rate increases the tendency for brittle fracture in steel.
2. Grain refinement increases the tendency for brittle fracture in steel.
3. High temperature increases the tendency for brittle fracture in steel.
4. Low temperature increases the tendency for brittle fracture in steel.

Question Number : 67 Question Id : 4410092079823 Question Type : MCQ Option Shuffling : No

Which region of the fatigue crack growth curve is typically associated with stable and predictable crack growth rates?

Options :

1. Region I
2. Region II
3. Region IV
4. Region III

Question Number : 68 Question Id : 4410092079835 Question Type : MCQ Option Shuffling : No

If the frequency used in eddy current testing is doubled, what happens to the depth of penetration?

Options :

1. The depth of penetration first increases then decreases
2. The depth of penetration increases
3. The depth of penetration remains the same
4. The depth of penetration decreases

Question Number : 69 Question Id : 4410092079818 Question Type : MCQ Option Shuffling : No

A manufacturing engineer needs to ensure a component resists ductile fracture under service loads. Which strategy is most effective?

Options :

1. Reducing the thickness of the component
2. Increasing the strain rate during operation
3. Lowering the service temperature of the component evenly
4. Selecting a material with high toughness and ductility

Question Number : 70 Question Id : 4410092079821 Question Type : MCQ Option Shuffling : No

A steel specimen undergoes fully reversed loading with a plastic strain amplitude of 0.02 and fails after 500 cycles. Which parameter is most directly assessed from this data using the Coffin-Manson relation?

Options :

1. Fracture toughness
2. Critical resolved shear stress
3. Elastic modulus
4. Fatigue ductility coefficient

Question Number : 71 Question Id : 4410092079822 Question Type : MCQ Option Shuffling : No

A steel specimen subjected to a stress amplitude of 250 MPa fails after 500,000 cycles. If the stress amplitude is reduced to 200 MPa, what is the expected trend in the number of cycles to failure?

Options :

1. The number of cycles to failure will decrease
2. The number of cycles to failure will remain unchanged
3. The number of cycles to failure will increase
4. The trend is typically linear with stress amplitude

Question Number : 72 Question Id : 4410092079824 Question Type : MCQ Option Shuffling : No

A fatigue crack grows from 1 millimeter to 4 millimeters under a constant stress intensity factor range. If the Paris Law exponent is 3, how does the crack growth rate change as the crack length increases?

Options :

1. Crack growth rate decreases by half
2. Crack growth rate remains unchanged
3. Crack growth rate increases eightfold
4. Crack growth rate doubles

Question Number : 73 Question Id : 4410092079826 Question Type : MCQ Option Shuffling : No

A cylindrical specimen is tested for creep at 900 Kelvin and 40 Mega Pascal with a steady-state creep rate of 1×10^{-7} per second. If the applied stress is doubled, and the creep exponent n equals 4, what is the new steady-state creep rate?

Options :

1. 1.6×10^{-6} per second
2. 2×10^{-6} per second
3. 8×10^{-7} per second
4. 4×10^{-7} per second

Question Number : 74 Question Id : 4410092079827 Question Type : MCQ Option Shuffling : No

During which stage of creep does necking and microstructural damage increase most rapidly, often leading to failure?

Options :

1. Tertiary creep stage
2. Secondary creep stage
3. Quaternary creep stage
4. Primary creep stage

Question Number : 75 Question Id : 4410092079829 Question Type : MCQ Option Shuffling : No

If the stress rupture strength of a material at 900 degrees Celsius is 100 mega pascals for a 1000 hour life, what is the expected rupture life at 80 mega pascals, assuming a power law relationship n equal to 4?

Options :

1. The rupture life increases by a factor of 10
2. The rupture life decreases by a factor of 0.5
3. The rupture life remains unchanged
4. The rupture life increases by a factor of 2.44

Question Number : 76 Question Id : 4410092079837 Question Type : MCQ Option Shuffling : No

Which of the following mathematical expressions correctly relates the magnetic field intensity H, magnetomotive force F, and length of the magnetic path l during the magnetic particle testing process?

Options :

1. $H = F + l$
2. $H = F / l$
3. $H = l / F$
4. $H = F * l$

Question Number : 77 Question Id : 4410092079856 Question Type : MCQ Option Shuffling : No

Which of the following defects is most commonly associated with improper forging temperature?

Options :

1. Lap
2. Cracking
3. Underfill
4. Scale

Question Number : 78 Question Id : 4410092079874 Question Type : MCQ Option Shuffling : No

Which of the following is a characteristic of powders produced by the electrolysis method?

Options :

1. High purity
2. Presence of oxides
3. Large particle size
4. Irregular shape

Question Number : 79 Question Id : 4410092079878 Question Type : MCQ Option Shuffling : No

Which parameter is most directly increased during the compaction process in powder metallurgy to achieve better green density?

Options :

1. Applied pressure
2. Average particle size
3. Lubricant addition
4. Sintering temperature

Question Number : 80 Question Id : 4410092079849 Question Type : MCQ Option Shuffling : No

Which of the following materials would exhibit the highest solidification shrinkage during casting?

Options :

1. Copper
2. Gray cast iron
3. Aluminum
4. Steel

Question Number : 81 Question Id : 4410092079868 Question Type : MCQ Option Shuffling : No

If the capillary action in a brazed joint is insufficient due to a large joint clearance, what is the most likely result?

Options :

1. Excessive flow of filler causing brittle joint
2. Incomplete filling and weak joint formation
3. Excessive diffusion between base and filler metals
4. Stronger bond due to easier filler penetration

Question Number : 82 Question Id : 4410092079872 Question Type : MCQ Option Shuffling : No

During welding, a student observes a crack propagating from the center of the weld bead towards the base metal. What type of defect is this?

Options :

1. Toe crack
2. Centerline crack
3. Overlap
4. Lamellar tearing

Question Number : 83 Question Id : 4410092079876 Question Type : MCQ Option Shuffling : No

Which property is most critical when selecting a powder for high-density sintered parts?

Options :

1. Magnetic permeability
2. Optical reflectivity
3. Particle compressibility
4. Electrical conductivity

Question Number : 84 Question Id : 4410092079845 Question Type : MCQ Option Shuffling : No

A riser with a volume of 800 cubic centimeters and surface area of 400 square centimeters is modified to have a volume of 900 cubic centimeters and surface area of 500 square centimeters. What is the percentage change in modulus?

Options :

1. Decrease of 10%
2. No change
3. Decrease of 12%
4. Increase of 12%

Question Number : 85 Question Id : 4410092079848 Question Type : MCQ Option Shuffling : No

If a foundry observes a recurring defect characterized by small pin-like holes on the surface of brass castings, which is the most probable cause?

Options :

1. High permeability during mold cavity preparation
2. Hydrogen absorption during melting
3. Excessive flux addition during furnace operation
4. Low pouring temperature during mold filling

Question Number : 86 Question Id : 4410092079858 Question Type : MCQ Option Shuffling : No

A billet is forged from 200 millimeters to 100 millimeters in height, using a constant true strain rate. What is the total true strain experienced by the billet?

Options :

1. 0.5
2. $\ln(100)$
3. $\ln(2)$
4. $\ln(0.5)$

Question Number : 87 Question Id : 4410092079865 Question Type : MCQ Option Shuffling : No

A circular blank of diameter 150 millimeter is drawn into a cup using a punch diameter of 90 millimeter. What is the drawing ratio?

Options :

1. 1.05
2. 0.60
3. 1.67
4. 1.15

Question Number : 88 Question Id : 4410092079871 Question Type : MCQ Option Shuffling : No

Evaluate the impact of substituting silver for lead in the tin-lead solder alloy on joint properties. Which statement is most accurate?

Options :

1. Lower wettability and decreased mechanical strength
2. Enhanced electrical conductivity and lower melting point
3. Improved mechanical strength and higher melting point
4. Reduced thermal fatigue resistance and brittleness

Question Number : 89 Question Id : 4410092079873 Question Type : MCQ Option Shuffling : No

Given a welding process where the voltage is 22 volts, current is 180 amperes, and travel speed is 6 millimeters per second, calculate the heat input per unit length. Choose the correct value.

Options :

1. 1320 joules per millimeter
2. 660 joules per millimeter
3. 396 joules per millimeter
4. 1080 joules per millimeter

Question Number : 90 Question Id : 4410092079881 Question Type : MCQ Option Shuffling : No

A powder compact with an initial porosity of 40% is sintered and achieves a final porosity of 10%. If the initial and final volumes are constant, what is the percentage reduction in pore volume?

Options :

1. 30%
2. 50%
3. 75%
4. 25%

Question Number : 91 Question Id : 4410092124332 Question Type : MCQ Option Shuffling : No

Which process is necessary to construct the adjoint of a 2x2 matrix?

Options :

1. Find the cofactors and transpose the resulting matrix
2. Multiply each entry by the determinant
3. Swap the first and second rows
4. Find the determinant and leave the matrix unchanged

Question Number : 92 Question Id : 4410092124331 Question Type : MCQ Option Shuffling : No

Suppose A is a 4x4 matrix with the property that $A^4 = 0$ but $A^3 \neq 0$. Which statement about the eigenvalues of A is correct?

Options :

1. A has four distinct eigenvalues.
2. Eigenvalues are all real and positive.
3. All eigenvalues of A are zero.
4. A has one nonzero eigenvalue.

Question Number : 93 Question Id : 4410091362119 Question Type : MCQ Option Shuffling : No

If a function is differentiable at $x = a$, then the tangent line at that point ____.

Options :

1. Always exists
2. Sometimes exists
3. Never exists
4. May be vertical

Question Number : 94 Question Id : 4410092124340 Question Type : MCQ Option Shuffling : No

What does the directional derivative of a scalar field at a point measure?

Options :

1. The rate of change of the field in a specified direction at that point
2. The total change of the field at all points
3. The magnitude of the gradient vector at that point
4. The rate of change of the field along the coordinate axes

Question Number : 95 Question Id : 4410092124345 Question Type : MCQ Option Shuffling : No

Given a vector field $F(x, y, z) = (2xy, x^2 + 2z, y^2)$, which statement best describes its irrotationality in R^3 ?

Options :

1. F is not irrotational in R^3
2. F is irrotational everywhere in R^3
3. F is irrotational only in the xy-plane
4. F is irrotational only on the plane $y=1$

Question Number : 96 Question Id : 4410091423662 Question Type : MCQ Option Shuffling : No

Which of the following differential equations is *separable*?

Options :

1. $(dy/dx) = x+y$
2. $(dy/dx) = xy$
3. $(dy/dx) = (x+y)/x$
4. $(dy/dx) = x+y^2$

Question Number : 97 Question Id : 4410092124352 Question Type : MCQ Option Shuffling : No

Consider the equation $\frac{dy}{dx} = x^3 + y^3$. What should be the first step in analyzing this equation?

Options :

1. Separate variables and integrate directly.
2. Classify the equation to determine whether it is linear, separable, exact, or requires a special solution technique.
3. Integrate both sides directly with respect to x.
4. Apply an integrating factor immediately.

Question Number : 98 Question Id : 4410092124360 Question Type : MCQ Option Shuffling : No

Which of the following is a requirement for applying Bayes theorem?

Options :

1. Only prior probabilities are needed
2. Events must be disjoint
3. Knowledge of conditional probabilities
4. Events must be independent

Question Number : 99 Question Id : 4410092124361 Question Type : MCQ Option Shuffling : No

A dataset contains the following points: (1,3), (2,5), (3,7), (4,9). If a straight line is fitted using the least squares method, which of the following best describes the resulting line?

Options :

1. The line will have a nonlinear relationship with the data points
2. The line will have a slope of approximately 2 and a y-intercept of approximately 1
3. The line will have a slope of approximately -2 and a y-intercept of approximately 3
4. The line will have a slope of approximately 1 and a y-intercept of approximately 2

Question Number : 100 Question Id : 4410092124366 Question Type : MCQ Option Shuffling : No

When applying Milne's method, what is the typical role of starting values?

Options :

1. They are not needed for Milne's method.
2. They are approximated using the Milne predictor formula only.
3. They are generated as the method progresses.
4. They are required to initialize the method using another method such as Runge-Kutta.